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Automatic Microstructural Classification of Ultrahigh Carbon Steel with Vision Transformers and Convolutional Neural Networks

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Abstract: Microstructural classification is key to better understanding of the relationships between the microstructures of carbon steels and their properties, as well as the development of automatic models to reliably and efficiently categorize the microstructures which can largely mitigate the burden of the human experts. Methods based on transfer learning with deep neural networks, such as convolutional neural networks (CNNs) have become established in automatic microstructural classification of carbon steels. However, vision transformers (ViTs), as a new class of deep learning algorithms, have recently emerged as a competitive alternative to CNNs in many other fields, but have not been considered in this field yet. In this study, the micrographs from an open-source dataset of ultrahigh carbon steel are analyzed by use of two CNNs (GoogLeNet and MobileNetV2) and two ViTs (ViT-B16 and ViT-L32). These deep neural networks are used as end-to-end classifiers to discriminate between different microstructures. Promising results were achieved by use of ViTs, as they performed as well as, or better than CNNs, though not by a large margin. These results suggest that ViTs can be a viable alternative to CNNs in the advancement of automatic microstructural classification.

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Keywords: vision transformer; texture analysis; convolutional neural networks; transfer learning; deep learning; microstructure; transformer; ultrahigh carbon steel.

1. INTRODUCTION

Carbon steel, an alloy of iron and carbon, is one of the primary categories of steel. It makes up as much as 90% of steel manufactured globally. With an increase in the carbon content, carbon steel can be divided into the following categories: ultralow carbon steel, low carbon steel, medium carbon steel, high carbon steel and ultrahigh carbon steel.

Ultrahigh carbon steel (UHCS) has approximately 1.25-2.0% carbon content. UHCS has extreme strength, sharpness, and resilience. It could be used for hard steel products, such as truck springs, metal cutting tools, and other special purpose items, such as knives, axles, or punches. UHCS can achieve extremely high hardness through special processing.

The microstructure of carbon steel is one of the critical factors determining its various properties, and mechanical behaviour in particular. Therefore, reliable classification of these microstructures is critical. Traditionally, classification is usually done by human experts. However, this way is inefficient and gives rise to uncertainties due to subjectivity.

Over the past several years, leveraging machine learning and deep learning, automatic classification of steel microstructures has been displacing traditional manual classification. Gola et al. (2019) have used support vector machine for the classification of combined morphological parameters and textural features for low carbon steel. Also in the same year, Arivazhagan et al. (2019) have reported that using a texture descriptor called modified alternate local ternary pattern, support vector machine outperformed k-nearest neighbours in the classification of three steel grades.

Convolutional neural networks (CNNs) are the most representative deep learning architectures for image processing tasks, which also captured the attention of researchers in steel microstructure classification. Azimi et al. (2018) have used fully convolutional neural networks accompanied by a max-voting scheme for microstructural classification in the examples of certain microstructural constituents of low carbon steel. They achieved nearly double the accuracy with the state-of-the-art method at that time. Following these early investigations, more researchers have used CNNs in the microstructural classification of carbon steel (Maemura et al., 2020; Jang et al., 2020; Muñoz-Rodenas et al., 2023a). These authors have used CNNs either for whole image classification or pixel-wise classification (Jang et al., 2020; Durmaz et al., 2021). It is noteworthy that most of these studies have focused on microstructure classification of low-carbon steel and that scant attention has been paid to UHCS to date.

Having made their first appearance in 2020 as a new class of deep learning architectures for image processing tasks, vision transformers (ViTs) (Dosovitskyi et al., 2021) are rapidly rising as a competitive alternative to CNNs. This is owing to their superior scalability to larger datasets, smaller image-specific inductive biases, higher computation efficiency, their ability to handle sequences of images, as well as being capable of interpreting structural dependencies between

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image features. An increasing number of areas have started using ViTs, e.g., face recognition, health sciences, and the process industries (Liu and Aldrich, 2023; Liu and Aldrich, 2024.). However, as far as the authors know, there have only been a few studies using ViTs in the microstructure classification of carbon steel, especially UHCS.

One such study conducted by Müller et al. (2021) was based on the use of a combination of Haralick, local binary pattern and morphological features as predictors in a support vector machine to classify seven different microstructures. Although reasonable results were obtained, the authors have indicated that there could be scope for improvement. In another, Muñoz-Rodenas et al. (2023b)

In a recent survey by Müller et al. (2024), the authors have likewise observed that convolutional neural networks and vision transformers are promising emerging applications in the classification of microstructures in steel, albeit that the relative merits of each approach in this context still had to be established.

Therefore, in this study, ViTs are used to classify different microstructures in UHCS, and are compared with CNNs. The results suggest that their performance can match or exceed that of CNNs and that ViTs are therefore a new class of deep learning methods that show exceptional promise in the development of automatic microstructural classification.

2. ANALYTICAL METHODOLOGY

2.1 Convolutional Neural Networks

GoogLeNet (Szegedy et al., 2015), a.k.a. Inception V1, developed by Google and its research partners, was the winner of the ILSVRC competition in 2014. Its architecture makes use of 1×1 or pointwise convolutions, global average pooling, as well as what is referred to as inception modules. As a result, it has a comparatively deep structure with relatively few parameters to train.

MobileNetV2 (Sandler et al., 2019) is a novel CNN architecture that has been adapted for use on mobile devices by significantly decreasing the number of operations and memory required, without sacrificing accuracy. This is achieved by using a novel layer module based on inverted residuals with linear bottlenecks taking a low-dimensional compressed representation. This representation is subsequently expanded to a high dimension and filtered using lightweight depth-wise convolution. Finally, linear convolution is used to project the features back to a low-dimensional representation.

2.2 Vision Transformers

With ViTs (Dosovitskyi et al., 2021), image classification problems are cast as sequence prediction tasks for series of image patches. This enables them to capture long-term dependencies within the input image. CNNs do not have this capability. In contrast, they learn to extract features hierarchically from images, among other owing to their use of progressively enlarged receptive fields.

ViT_B16 is a base variant of vision transformer models while ViT_L32 is a large-scale variant. Both models are pretrained on the ImageNet-21k dataset and finetuned on the ImageNet-1k dataset. By use of the transfer learning mode, the input images were resized to 224 × 224 resolution for ViT_B16 model and 384 × 384 resolution for ViT_L32 model. ViT_B16 model has 12 transformer layers, 86.86 million trained parameters and its patch size is 16 × 16. ViT_L32 model has 24 transformer layers, 305.55 million trained parameters, and its patch size is 32 × 32. As can be seen from the number of trained parameters and number of layers, ViT_L32 is more powerful in terms of processing capability and potential performance, however, with the requirement of more computational resources. From the difference in patch size which affects how the model understands the content of the image, ViT_B16 may be better at capturing details while ViT_L32 may be better at understanding a broader context of the image.

3. CASE STUDY

In the case study, the public Ultrahigh Carbon Steel Micrograph DataBase (UHCSDB) of 961 scanning electron microscopy (SEM) UHCS micrographs (DeCost et al., 2017) was considered to classify the textures. The seven classes of primary microconstituents are: Class A (martensite), Class B (pearlite), Class C (pearlite + spheroidite), Class D (pearlite + Widmanstätten), Class E (proeutectoid cementite network), Class F (spheroidite) and Class G (spheroidite + Widmanstätten). Examples from each class are shown in Fig. 1.

The associated number of each class is indicated just after the microconstituent in the parenthesis.

Two CNNs and another two ViTs were used as end-to-end classifiers. The CNNs used in this investigation were built using a PyTorch backend, while the ViTs were built on a TensorFlow backend. All the experiments were run on a single graphics processing unit (GPU) device on the Google Colab platform.

As suggested in a previous study by the authors (Liu and Aldrich, 2022), transfer learning strategy makes a marked difference, even when dealing with small data sets, and full retraining is recommended when sufficient computational resources are available. Therefore, full retraining or fine-tuning was used, that is, all the trainable parameters in the CNNs and ViTs were updated during the training process.

During retraining of the CNNs or ViTs, images of the micrographs were randomly split into training and test data sets in a ratio of 8:2, with the latter used as an independent test set to validate the generalization of the deep learning models. The training set was further randomly shuffled, and 75 % of it was used to train the models, while the remaining 25 % was allocated to a validation image data set. After training was completed, the models were tested with the test data set not used during training and validation of the models.

The classification performance of the four different models is summarized in Table 1, together with the dimension of the

corresponding feature set. The classification performance was based on the percentage of images correctly classified.

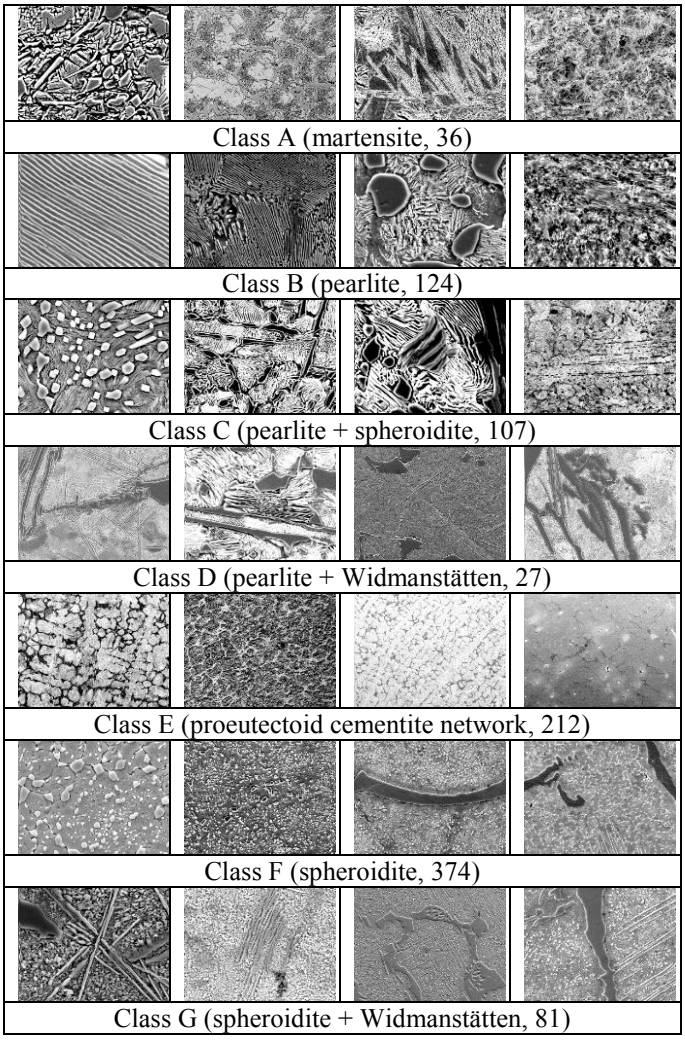


Fig. 1. Typical images of microstructures in UHCS associated with Classes A-G.

As can be seen in Table 1, all the four deep learning models performed quite well, achieving relatively high accuracies of 87.2~91.3%. Of the four, ViT_L32 is the best model, though just by a small margin 1~1.4% compared to the two CNN models. For the two ViT models, it is obvious that the larger model is performing better than base model in this case study, with an improvement of accuracy by 4.1%.

Table 1. Performance comparison between different models.

Models	Number of Features	Accuracy (%) Test Data
ViT_B16	768	87.2
MobileNetV2	1280	89.8
GoogLeNet	1024	90.3
ViT_L32	1024	91.3

However, the training of large ViT models requires GPUs with higher VRAM. For example, in this case study, the ViT_L32 model can only be run on an A100 GPU with 40 GB VRAM, while the ViT_B16 model can be executed on a V100 GPU with less VRAM of 16 GB.

The discriminative power of the different models can be further assessed by visualising the features extracted from the images in a two-dimensional score space by use of a *t*-distributed stochastic neighbour embedding (*t*-SNE) score plot. The features were extracted from the "ExtractToken" layer for the ViT models, as well as the layer immediately preceding the last fully connected layer for the CNN models.

Figs 2 and 3 show the *t*-SNE score plots of the features extracted from the different models with the corresponding classification accuracy. In these graphs, the seven classes, Class A-G are respectively represented by 'black dot', 'red plus', 'light blue asterisk', 'dark blue star', 'green triangle', 'magenta diamond' and 'yellow circle' markers. As can be seen from Figs 2 and 3, all four feature sets can separate the seven classes satisfactorily.

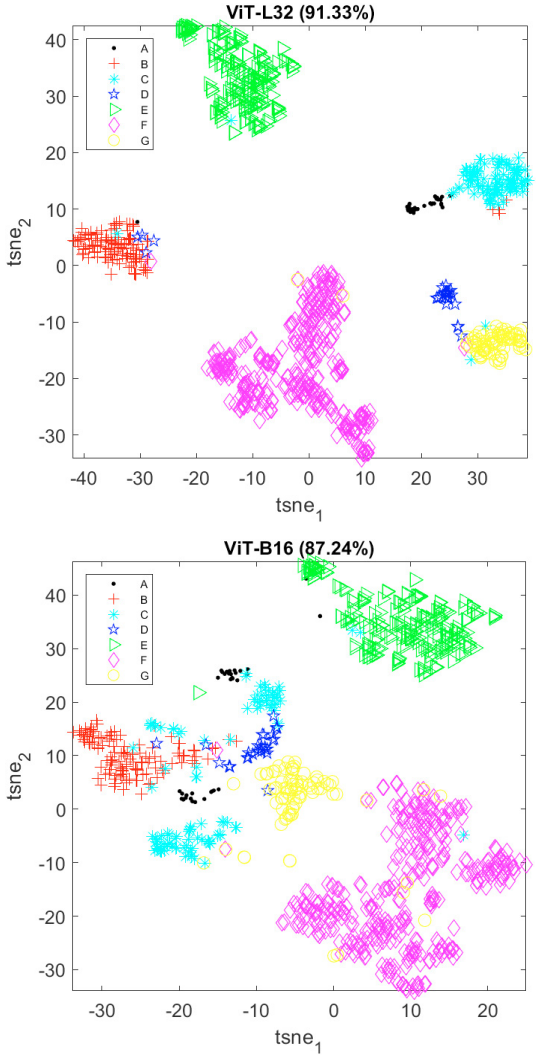


Fig. 2. t-SNE score plot of the features extracted from the ViT models.

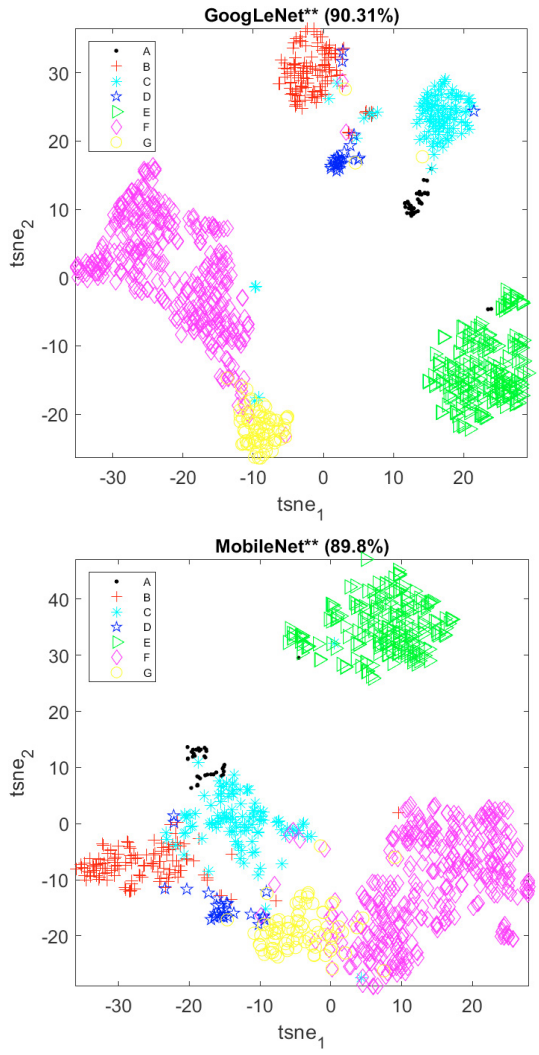


Fig. 3. t-SNE score plot of the features extracted from the CNN models.

To further demonstrate the classification performance of the best model ViT_L32, the confusion matrix on the test set is shown in Fig. 4. There are 6-75 samples for each class in the test set. The actual observations are presented in rows, while the predicted labels are presented in the columns of the table. More specifically, the numbers on the diagonal represent correct predictions, while the off-diagonal numbers give insight in the failures of the model.

As can been seen from Fig. 4, most of the seven classes (except Class D) can be distinguished well from other classes, especially Class E can be distinguished from the other classes perfectly. This is consistent with the visualization results in the corresponding *t*-SNE plot. This may in part be owing to the small dataset available, class imbalance, and the statistical variation in the images themselves, which could have contributed to the intractability of the problem.

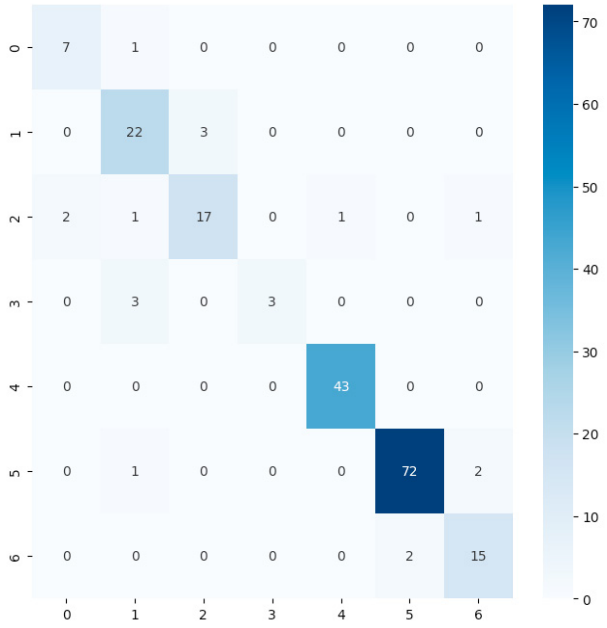


Fig. 4. Confusion matrix on test set for ViT_L32.

6. CONCLUSIONS

In this study, the automatic microstructural classification of ultrahigh carbon steel was implemented with two models from a new class of deep neural networks, vision transformers. Through the comparison with another two models from convolutional neural networks, we found that these two classes of deep neural networks can perform as well as each other. Furthermore, using larger ViT model can achieve slightly accuracy. This suggests that ViTs can be a competitive alternative to CNNs in this task.

However, which ViT variant should be adopted needs to take some factors into account, such as the availability of computational resources and the volume of available training data. Base ViT models may be more suitable for resource-constrained or data-limited scenarios, while large ViT models are more suitable for large-scale datasets and high-performance computing environments. There are also huge ViT models available, but they are more applicable to huge datasets correspondingly and require much more computational resources.

In the future, the class imbalance problem might be mitigated by image synthesis to enlarge the dataset. As suggested by previous works by the authors (Liu and Aldrich, 2022; Liu and Aldrich 2023), image synthesis can help to improve the model performance. Although this is computationally expensive, a first feasible try might be only generating synthetic micrographs for the minority classes, like Class D.

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